

Retail Sales Data Warehouse

Stage Layer

roqiua mohammad 10/15/26

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Business Reequipments:

The staging layer aims to consolidate raw data from the operational SQL Server database and the external product catalog file into a centralized area before loading it into the data warehouse. It should preserve source data and identifiers while supporting data validation and quality checks.

Task Requirements

- Create five staging tables under the stg schema in the RetailDWH database.
- Extract data from the four source tables in RetailOLTP.
- Load product data from ProductCatalog_Batch1.csv.
- Preserve source columns, business IDs, and modification timestamps.
- Implement a full-refresh loading process.
- Ensure the staging layer is ready for subsequent ETL processing.

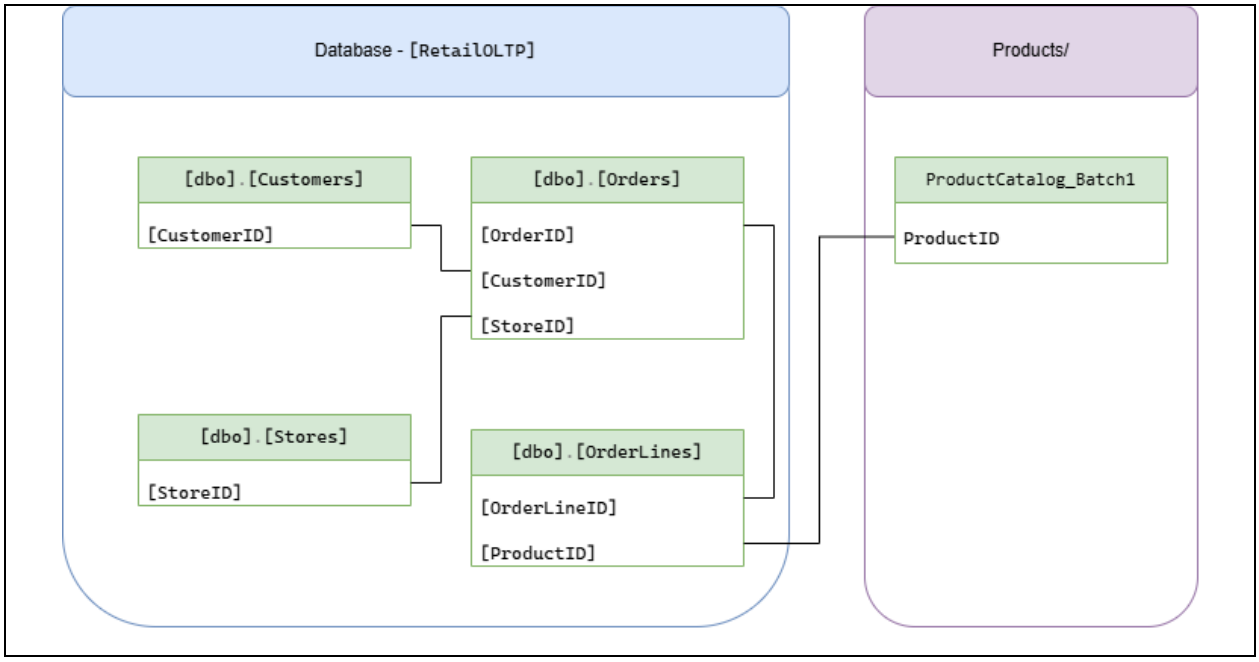
Data Sources:

Total: 5 Sources

1. [dbo].[Customers]	
Source Type	Database – Sql Server
Source Location	[RetailOLTP]
Description	Customer master data, including names and contact details.
2. [dbo].[Orders]	
Source Type	Database – Sql Server
Source Location	[RetailOLTP]
Description	Sales order header information, including customer, store, order date, and status.
3. [dbo].[OrderLines]	
Source Type	Database – Sql Server
Source Location	[RetailOLTP]

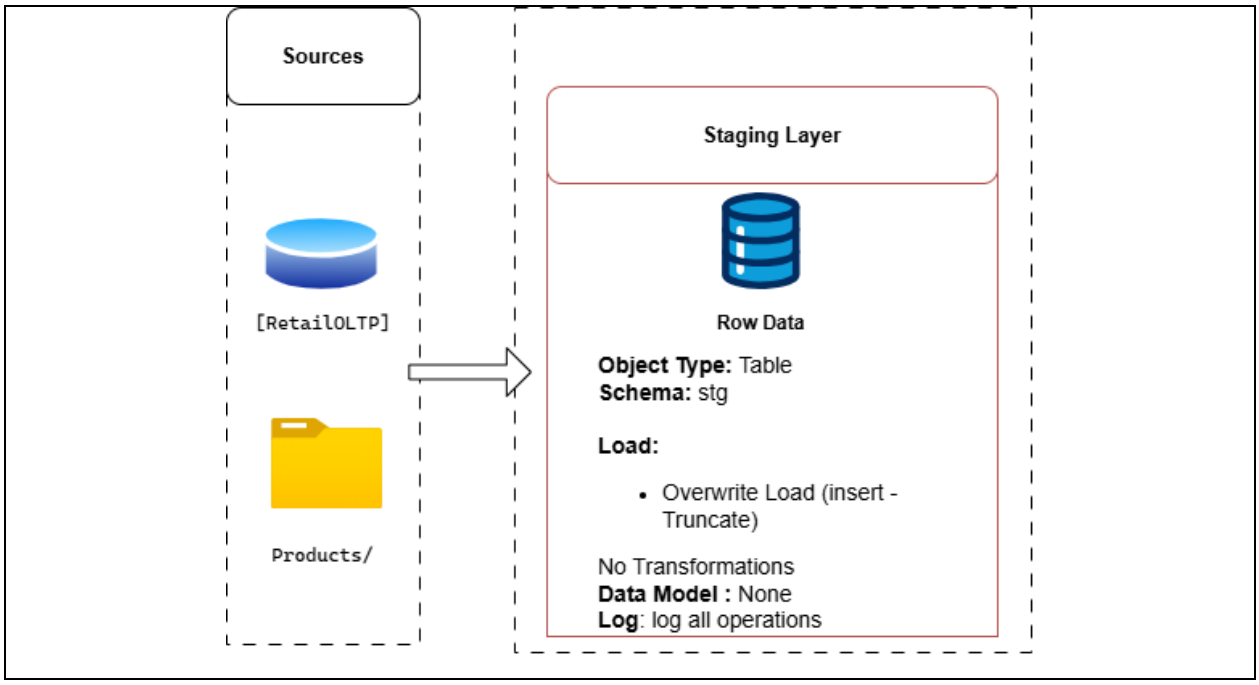
Description	Sales transaction details, including products, quantities, prices, and discounts.
4. [dbo].[Stores]	
Source Type	Database – Sql Server
Source Location	[RetailOLTP]
Description	Store master data, including store names and locations.
5. ProductCatalog_Batch1	
Source Type	Excel
Source Location	/Products
Description	Sales transaction details, including products, quantities, prices, and discounts.

Data Integrations (How tables are related):

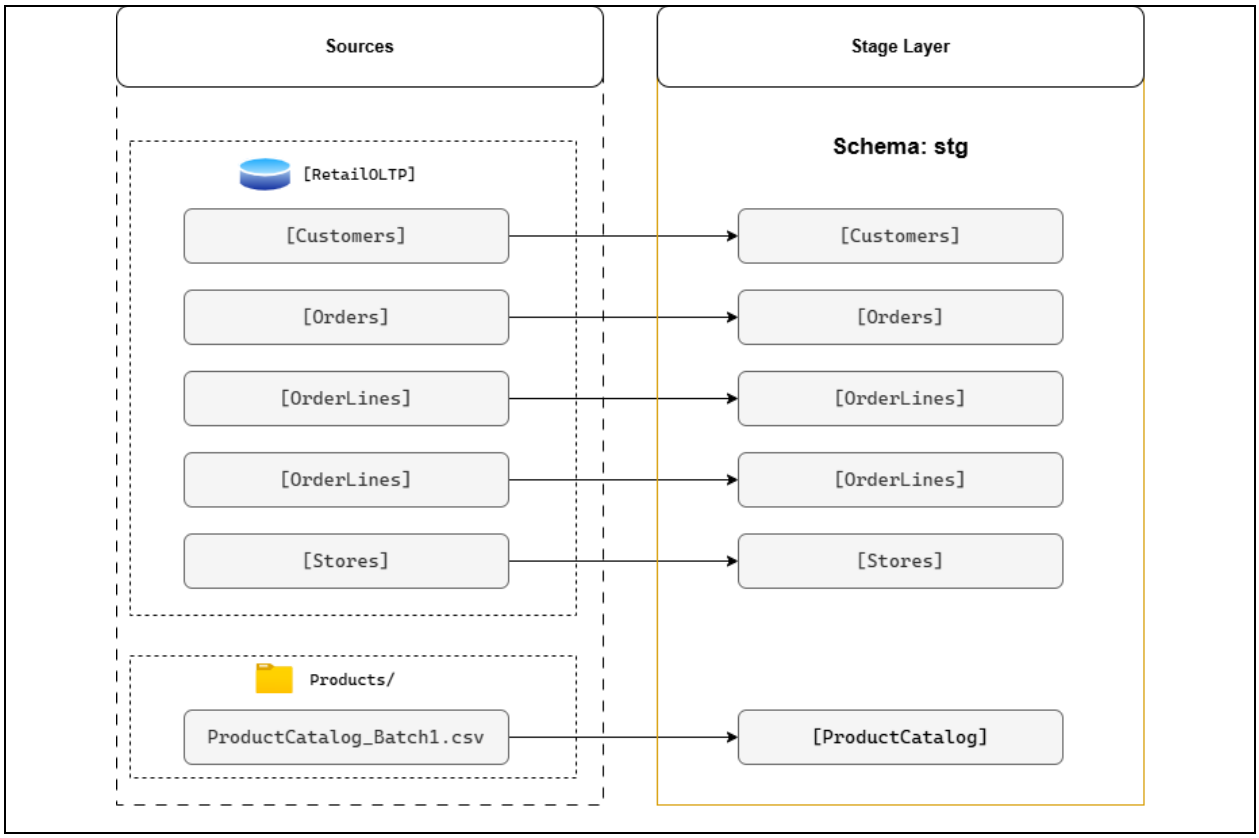


Data Architecture:

- Architecture Diagram:



- Data Flow:



ETL Pipeline:

1. Pipeline Structure

A master SSIS package is executed, containing a separate child package for each staging table. Each child package is responsible for loading data into its corresponding staging table.

2. Batch Tracking

Each master pipeline execution creates a new batch record containing the batch start and end timestamps. The batch status is determined based on the execution results of all child packages.

3. Table Loading Process

Each child package follows these steps:

1. **Truncate:** Execute the stored procedure stg.truncate_table_stage_layer to truncate the target staging table.
2. **Log Truncate Operation:** Record the truncate operation and check whether it succeeded or failed.
3. **Insert Data:** If the truncate operation succeeds, insert the source data into the staging table.

4. **Log Insert Operation:** Record the insert operation and its execution result.

If the truncate operation fails, the failure is logged, and the insert operation is skipped.

4. Error Handling

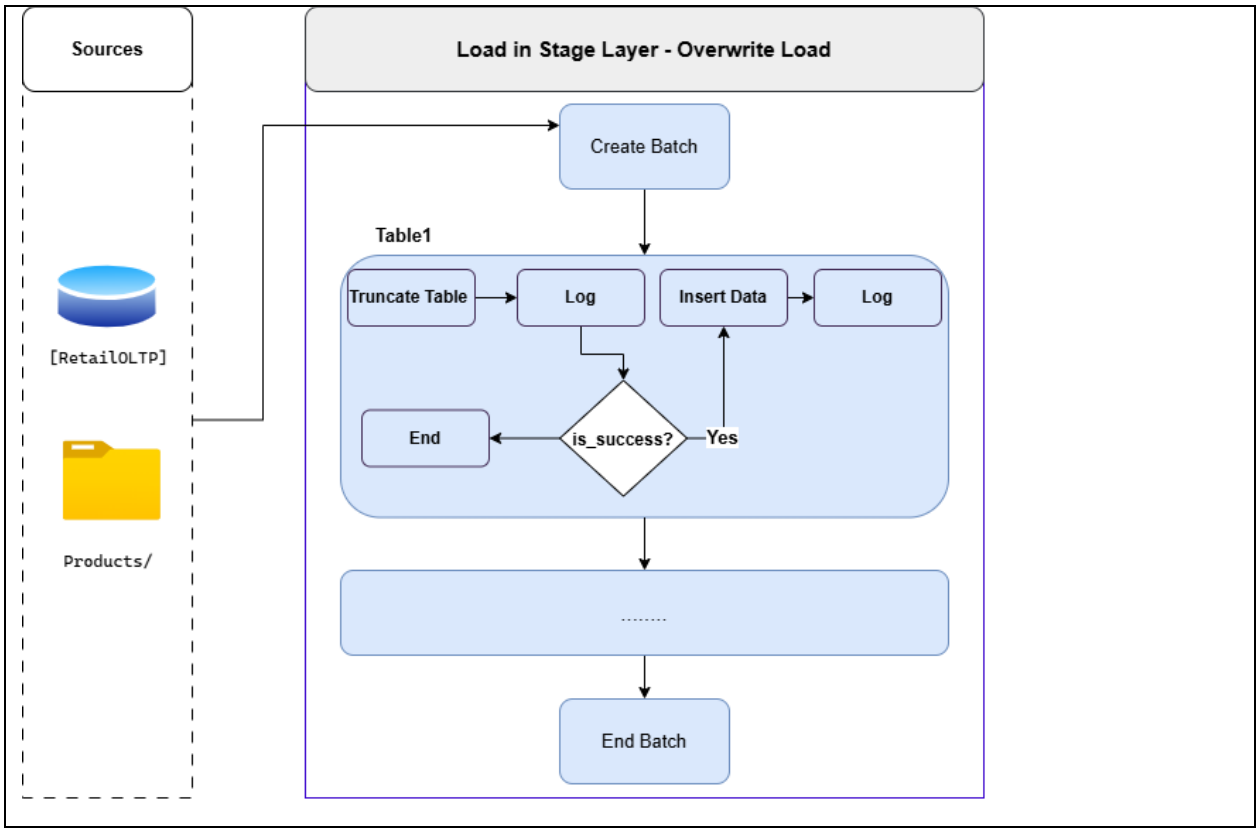
If a child package fails, the master package continues executing the remaining child packages independently. A failure in one package does not prevent the other packages from running.

5. Batch Status

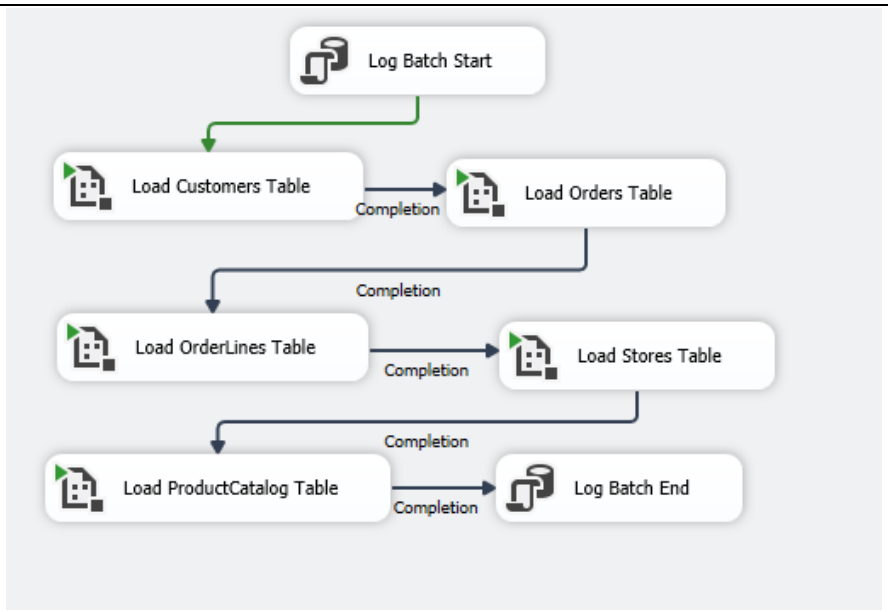
- **Succeeded:** All child packages complete their required data transfer operations successfully.
- **Failed:** At least one child package encounters a failure in a data transfer operation.

6. DML Operations

The staging data loading process uses only TRUNCATE and INSERT operations. Logging and batch tracking are handled separately to record execution details and outcomes.

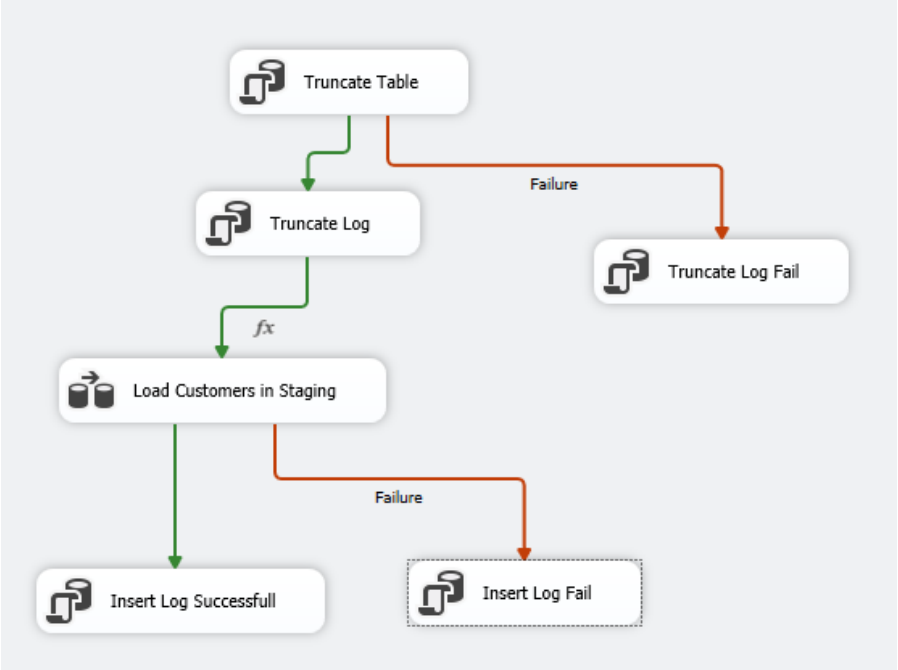


Execute Package:

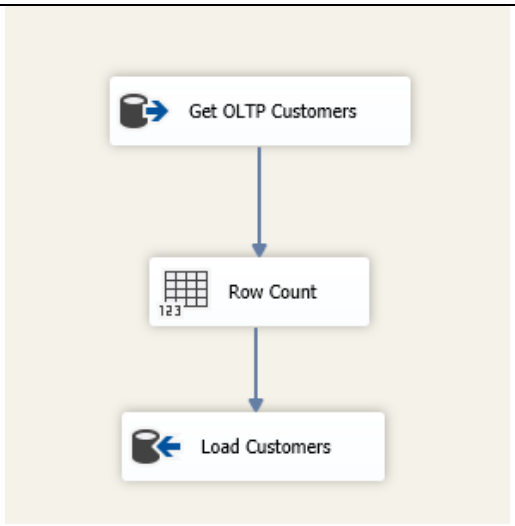


Example Table Package

Control Flow



Data Flow



Example Log Batch:

Batch

	batch_id	start_date	end_date	status
1	1	2026-10-17 07:31:53.3233333	2026-10-17 07:32:32.2266667	Succeeded

Log operations in Batch

Results | Messages

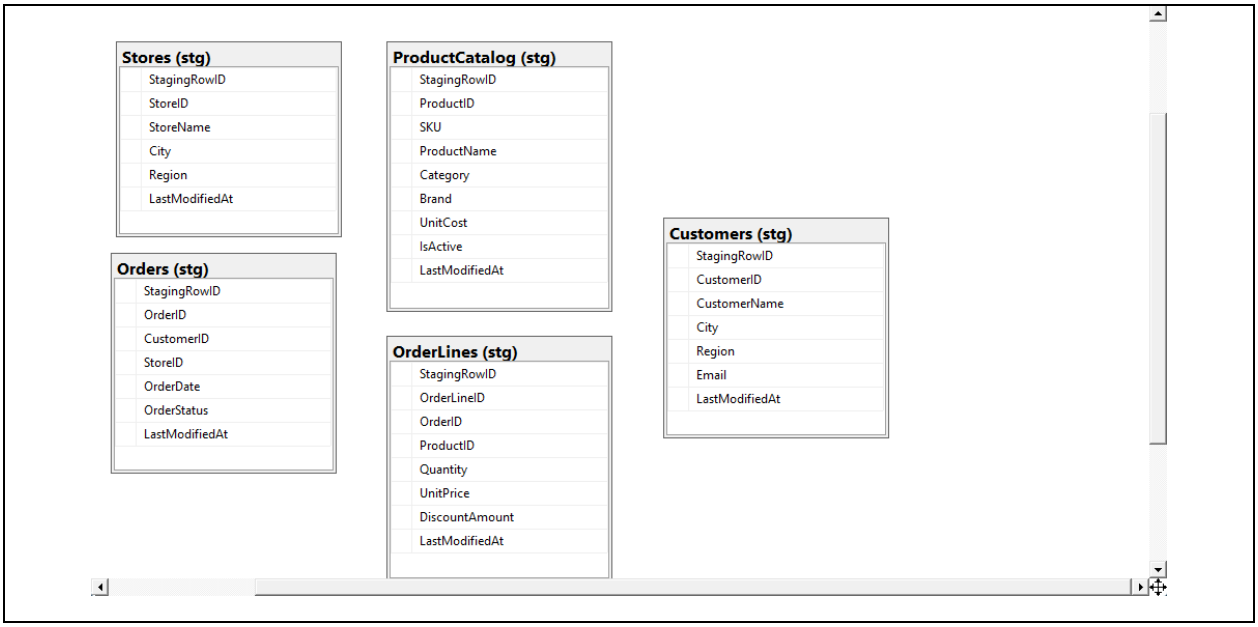
	log_id	batch_id	table_name	operation_type	load_type	no_rows	is_success	message	execution_date
1	1	1	Customers	truncate	Overwrite	0	1	Truncate completed successfully.	2026-10-17 07:31:55.393
2	2	1	Customers	insert	Overwrite	30	1	Insert completed successfully.	2026-10-17 07:32:00.403
3	3	1	Orders	truncate	Overwrite	0	1	Truncate completed successfully.	2026-10-17 07:32:02.990
4	4	1	Orders	insert	Overwrite	180	1	Insert completed successfully.	2026-10-17 07:32:10.693
5	5	1	OrderLines	truncate	Overwrite	0	1	Truncate completed successfully.	2026-10-17 07:32:11.587
6	6	1	OrderLines	insert	Overwrite	470	1	Insert completed successfully.	2026-10-17 07:32:20.737
7	7	1	Stores	truncate	Overwrite	0	1	Truncate completed successfully.	2026-10-17 07:32:21.257
8	8	1	Stores	insert	Overwrite	6	1	Insert completed successfully.	2026-10-17 07:32:30.323
9	9	1	ProductCatalog	truncate	Overwrite	0	1	Truncate completed successfully.	2026-10-17 07:32:30.937
10	10	1	ProductCatalog	insert	Overwrite	24	1	Insert completed successfully.	2026-10-17 07:32:32.143

Data Layers:

- Stage Layer:
  - Data Dictionary:

- **Data Loading:** Data is transferred from the source systems to the staging tables without any modifications or transformations.
- **StagingRowID**: An IDENTITY column is added to each staging table as a technical identifier for staging records only. It is not a surrogate key used in the data warehouse dimensions.

○ Tables:



Technologies Used:

- **SQL Server:** Database management and data warehouse implementation.
- **T-SQL:** Data cleaning, transformation, business rules, and stored procedures.
- **SQL Server Management Studio (SSMS):** Database development, query execution, and data validation.
- **SQL Server Integration Services (SSIS):** ETL package development, orchestration, staging table loading, and error handling.

Links & References:

- **LinkedIn:** www.linkedin.com/in/roqiua-mohammad
- **GitHub Repository:** https://github.com/roqiua/RetailDWH_bootcamp_databill